

BUSINESS SYSTEMS ANALYSIS (BSA)

GROUP PROJECT – MASTER CASE SCENARIOS & DOMAIN ALLOCATION

PART 1. PROJECT CASE ASSIGNMENT & DOMAIN ALLOCATION

Each group is assigned a **Master Business Scenario** paired with a **Specific Business Context**. While multiple groups share the same broad scenario, each group must analyze a unique industry domain.

Students are expected to investigate their assigned domain, identify stakeholders and business needs, model AS-IS and TO-BE processes, define system requirements (BRD), and validate the proposed solution through system configuration and UAT testing.

An **Odoo ERP environment** will be provided as a platform for solution evaluation and prototyping.

CRITICAL RULE: Students must determine which system functionalities are appropriate based on their business analysis and requirements, **rather than simply implementing all available Odoo modules default features.**

Group	Business Scenario / Domain	Specific Business Context	Key Business Characteristics & Master Data Requirements
1	Scenario A: Omnichannel Retail	Fashion & Apparel Chain	Showroom sales, e-commerce, inventory sync, POS. <i>Data: Size/Color variants, seasonal items.</i>
2	Scenario A: Omnichannel Retail	Consumer Electronics Store	Serialized products, warranty management, customer invoicing. <i>Data: Serial/IMEI tracking.</i>
3	Scenario A: Omnichannel Retail	Home Decor & Furniture	Delivery scheduling, stock availability, down payments. <i>Data: Custom shipping fees, deposit rules.</i>
4	Scenario A: Omnichannel Retail	Cosmetics & Beauty Chain	Expiration dates, multi-location stock, loyalty sales. <i>Data: Batch/Lot numbers, expiry tracking.</i>
5	Scenario B: B2B Wholesale	FMCG Distribution	Bulk discounts, vendor POs, credit limits. <i>Data: Tiered pricing lists, customer credit thresholds.</i>
6	Scenario B: B2B Wholesale	Office Supplies & Stationery	Contract pricing, purchase requisitions, customer invoicing. <i>Data: Contract-based price agreements.</i>

7	Scenario B: B2B Wholesale	Industrial Spare Parts	Reordering rules, lead time, multi-vendor POs. <i>Data: Min/Max stock reorder points, vendor lead time.</i>
8	Scenario C: Light Manufacturing	Custom Gift Hamper Assembly	Component procurement, kitting, finished goods stock. <i>Data: Kitting Bill of Materials (BOM).</i>
9	Scenario C: Light Manufacturing	Bicycle & Scooter Assembly	Bill of Materials (BOM), assembly stock, sales orders. <i>Data: Multi-level sub-assemblies/parts.</i>
10	Scenario C: Light Manufacturing	Coffee Roasting & Packaging	Raw beans purchasing, batch processing, B2B sales. <i>Data: Batch yields, raw vs. finished goods stock.</i>

PART 2. PROJECT SCOPE & BOUNDARIES

To prevent scope creep and maintain academic rigor, all projects must adhere strictly to the following boundaries:

- **Core Functional Scope (Max 4 Modules):** All groups are constrained to analyzing and configuring **Sales, Purchase, Inventory, and Invoicing/Accounting**.
- **Master Data Rule:** Each group must define at least **5–10 realistic SKUs/Products** with attributes specific to their domain context (e.g., variants, serial numbers, expiration dates, or BOMs).
- **AI-Assisted Stakeholder Interviewing:** Students will conduct requirement-gathering interviews using pre-configured **AI Stakeholder Agents** (accessible via Google AI Studio links provided on LMS).

SCENARIO A: OMNICHANNEL RETAIL

Applicable groups: 1, 2, 3, and 4

1. Enterprise Background

Company A operates a physical showroom and an online store. The business is growing, but several operational activities are managed using separate systems and manual procedures.

Showroom sales staff use an offline POS system, while online orders are frequently managed through messaging applications and spreadsheets. Different departments maintain their own records, making it difficult to obtain consistent and timely information across the organization.

Management is concerned that the current way of working may not adequately support the company's continued growth.

2. Initial Business Problems and AS-IS Challenges

The following issues have been reported by management and employees:

Sales

Sales representatives sometimes confirm online customer orders without reliable information about current product availability. This may result in order cancellations, delayed deliveries, or additional communication with customers.

Information about customer deposits and payment status is not always clearly shared between Sales and Accounting.

Purchasing

The purchasing team generally initiates replenishment when products are already low or unavailable. This can create periods in which popular products cannot be supplied to customers.

Inventory

Showroom inventory and central warehouse inventory are maintained separately. Some stock movements between locations are recorded manually, and recorded quantities do not always match physical stock.

Invoicing

Customer invoices are often prepared manually after sales transactions. Accounting also spends time matching customer payments with the corresponding sales transactions.

3. Project Investigation

Based on the information provided above, the project team should investigate:

- How the business currently operates
- Who the key stakeholders are
- What information each stakeholder needs
- Which business processes are affected
- Where the main problems and bottlenecks occur
- What business requirements should be addressed
- What functional requirements may be needed
- Which system capabilities could potentially support the requirements
- How the proposed solution could be validated

The project team should **not assume that every identified problem requires a new system function**. Proposed solutions should be justified based on the analysis.

4. Available System Environment

The project team will have access to an Odoo-based enterprise system environment that may provide capabilities related to areas such as:

- Sales
- Purchase
- Inventory
- Invoicing
- Customer and product information
- Stock locations and movements
- Pricing and payment management

Students should identify and justify the system capabilities that are relevant to their requirements.

SCENARIO B: B2B WHOLESALE & DISTRIBUTION

Applicable groups: 5, 6, and 7

1. Enterprise Background

Company B is a regional B2B distributor supplying products to corporate clients, supermarkets, and sub-dealers.

The company handles relatively large orders and works with different suppliers and customer groups. Customers may receive different pricing conditions and payment terms depending on their business relationship with the company.

As the business grows, employees report increasing difficulties in coordinating Sales, Purchasing, Inventory, and Accounting activities.

2. Initial Business Problems and AS-IS Challenges

Sales

Sales representatives negotiate different pricing and discount conditions with B2B customers. These arrangements may be communicated through emails or informal discussions, making it difficult to maintain consistent pricing information.

Sales representatives may also accept orders from customers with overdue balances, creating concerns for Accounting and management.

Purchasing

The purchasing team works with multiple local and international suppliers. Information about supplier quotations, purchasing prices, delivery lead times, and incoming shipments is maintained across different sources.

This makes supplier comparison and purchasing follow-up time-consuming.

Inventory

Incoming goods may remain at the receiving area before being formally checked and recorded. As a result, Sales and Warehouse staff may not have timely information about newly received products.

Invoicing and Payments

Many customers receive credit terms, such as 30-day payment terms.

Accounting spends considerable time monitoring payment due dates and matching partial payments with outstanding customer invoices.

3. Project Investigation

Based on the initial information, the project team should investigate:

- How B2B sales and purchasing activities currently operate
- Who is involved in pricing, credit approval, purchasing, receiving, and payment processes
- What information is required at each stage
- Where process bottlenecks and information gaps occur
- Which business problems should be prioritized
- What business and functional requirements should be defined
- Which system capabilities could support the requirements
- How the proposed solution could be evaluated and validated

Students should justify the proposed solution based on the identified business needs and requirements.

4. Available System Environment

The project team will have access to an Odoo-based enterprise system environment that may provide capabilities related to:

- Sales
- Purchase
- Inventory
- Invoicing
- Customer and vendor management
- Pricing and discount management
- Payment terms and payment tracking
- Stock receipts and transfers

Students should determine which capabilities are relevant to the identified requirements and provide justification for their selection.

SCENARIO C: LIGHT MANUFACTURING & KITTING

Applicable groups: 8, 9, and 10

1. Enterprise Background

Company C purchases raw materials, parts, or individual components from various suppliers and assembles or packages them into finished products for sale to B2B and B2C customers.

The company needs to coordinate Sales, Purchasing, Inventory, and Assembly activities. As the number of products and orders increases, employees experience difficulties in maintaining accurate information about component availability, production or assembly status, and product costs.

2. Initial Business Problems and AS-IS Challenges

Sales

Sales representatives sometimes accept orders for customized products or bundles without having complete information about the availability of the required components.

This can result in delays when the company cannot obtain all required components in time.

Purchasing

Component requirements are frequently calculated manually. Purchasing staff may order excessive quantities of some components while failing to identify shortages of critical components.

Inventory

The company needs to manage both component stock and finished-product stock.

When components are used during assembly, the corresponding inventory records are not always updated consistently.

Costing and Invoicing

The cost of finished products can vary because component purchase prices change over time.

Accounting therefore has difficulty determining accurate product costs and analyzing profit margins.

3. Project Investigation

Based on the initial information, the project team should investigate:

- How components move through the organization
- How sales orders affect purchasing and assembly activities
- Which stakeholders are involved in planning, purchasing, inventory, assembly, and sales
- What information is required to make decisions
- How the current processes create delays, errors, or information gaps
- Which business requirements should be prioritized
- What functional requirements are needed
- Which system capabilities could support the requirements
- How the proposed solution could be evaluated using the available system environment

Students should distinguish between **business needs/problems** and **system requirements**, and should justify how the proposed solution addresses the identified problems.

4. Available System Environment

The project team will have access to an Odoo-based enterprise system environment that may provide capabilities related to:

- Sales
- Purchase
- Inventory
- Product and component management
- Bill of Materials (BOM)
- Manufacturing or assembly activities
- Invoicing
- Vendor and customer management

Students should determine which capabilities are appropriate for the identified requirements and should not assume that all available capabilities must be used.

PART 3. PROJECT PRINCIPLES

Regardless of the assigned scenario, all groups should follow these principles.

1. Analyze before proposing a solution

Students should not begin with the question:

“Which Odoo module should we use?”

Instead, begin with:

“What is the business problem, who is affected, and what does the organization need?”

2. The solution must be traceable to the business need

The project should demonstrate the following relationship:

**Business Problem → Process/Pain Point → Business Need → Requirement →
System Functionality → Validation**

Students should be able to explain why each major system functionality is included in the proposed solution.

3. Odoo is a solution environment, not the focus of the course

Odoo is provided to allow students to:

- evaluate whether proposed requirements can be supported;
- explore possible system solutions;
- prototype selected functionalities;
- demonstrate how requirements can be translated into system capabilities; and
- validate the proposed solution.

The project is primarily a **Business Systems Analysis project**, not an Odoo training project.

4. Roles represent different perspectives

During the project, students may work in different roles, such as:

- **Business Owner / Stakeholder**
- **Business Systems Analyst**
- **Project Manager**
- **Developer / System Configurator**
- **Tester / User Acceptance Tester**

Roles may rotate across project phases.

A student working as a Developer, Tester, Project Manager, or Business Owner is still expected to participate in the required business analysis activities and demonstrate an understanding of how business needs are translated into system requirements.

5. Individual and group learning

Each student is expected to contribute to both:

Group outputs

- Business analysis
- Process models
- Requirements documentation
- Use cases
- Solution prototype
- Testing and validation
- Final presentation

Individual evidence

- Stakeholder questions/interview activities
- Problem analysis
- Process analysis
- Requirement contributions
- Use case contributions
- Validation/UAT activities
- Individual reflection and presentation

The group project demonstrates the team's ability to integrate the analysis and solution, while individual activities demonstrate each student's learning and contribution.

PART 4. PROJECT PHASES

The project will be developed progressively rather than completed as one final assignment.

Phase 1 – Business Analysis

Students investigate the assigned organization and develop an understanding of:

Business → Problems → Stakeholders → Processes → Information Needs

Expected outputs include:

- Enterprise profile
 - Business context
 - Stakeholder identification
 - Business problem analysis
 - Information needs
 - AS-IS process models
 - Pain-point analysis
 - TO-BE process proposals
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Phase 2 – Systems Analysis and Solution Evaluation

Students transform the results of Phase 1 into:

Requirements → Use Cases → Solution → Prototype → Validation

Expected outputs include:

- Business requirements
- Functional requirements

- System scope
- Actors
- Use case diagram
- Use case descriptions
- Requirement-to-process/solution mapping
- Selected Odoo configuration or prototype
- Test/UAT scenarios
- Validation results
- Final recommendations

The final solution should demonstrate a clear connection between the organization's business problems, the identified requirements, and the proposed system functionality.

PART 5. ASSESSMENT & SUBMISSION MILESTONES

Milestone	Weeks	Primary Activities & Outputs	Submission Deliverables
Phase 1: Mid-Term Assessment <i>(Weight: 30%)</i>	Weeks 1–4	• Domain Setup & Group RACI Assignment Matrix. • Conduct AI Stakeholder Interviews & Log Verified Evidence.	Draft BRD (Sections 1 & 2)
	Weeks 5–9	• AS-IS Process Modeling & Bottleneck Identification. • Root Cause Analysis (Fishbone / 5-Whys). • TO-BE Process Modeling & MoSCoW Requirements List.	Phase 1 BRD Report <i>(Sections 1, 2, and 3)</i>
Phase 2: Final BRD Assessment <i>(Weight: 40%)</i>	Weeks 10–11	• Enterprise Use Case Diagram. • Detailed Use Case Specifications (per Module). • Business Rules & Operational Constraints Matrix.	Section 4 Specifications

	Weeks 12–13	• Odoo ERP Core Setup & Domain SKU Master Data. • Design & Execute Peer UAT Testing. • Document Executed Test Log with Pass/Fail Outcomes.	Complete Master BRD <i>(Sections 1–4 + Appendices A, B, C)</i>
	Weeks 14–15	• Team Slide Presentation. • Live End-to-End Odoo Workflow Demonstration. • Individual Q&A Defense before Instructors.	Presentation Slides & Live Demo

Phase	Weeks	Strategic Focus	Primary Student Activities & Output	Key Deliverables & Output Artifacts
Phase 1: Business Analysis & Requirements Engineering <i>(30% Weight)</i>	Weeks 1 – 2	Domain Setup & Context Definition	Assign groups to Master Case Scenarios (Groups 1–10) and define project timelines/RACI matrix.	Section 1 & 1.4: Enterprise Context & RACI Assignment Matrix
	Weeks 3 – 4	Stakeholder Elicitation	Conduct structured interviews with pre-configured AI Stakeholder Agents in Google AI Studio.	Section 2 & Appendix: Information Needs Matrix & Chat Transcripts Log
	Weeks 5 – 7	Business Process Modeling	Analyze operational gaps and design current (AS-IS) and future (TO-BE) workflow diagrams.	Section 3 & 2.x.1: End-to-End & Departmental BPMN Diagrams (AS-IS & TO-BE)

	Weeks 8 – 9	Requirements Framing	Translate process gaps into prioritized functional requirements using the MoSCoW framework.	Section 2.x.2: Prioritized Functional Requirements (FRs) List → BRD Phase 1 Submission
Phase 2: Solution Mapping, Odoo Config & UAT (40% Weight)	Week 10	Detailed Functional Specification	Select core functional requirements and elaborate on operational scenarios and exception paths.	Section 4: Detailed Use Case Specifications (Basic, Alternate & Exception Flows)
	Week 11	Quality Assurance & Alignment	Perform peer reviews on specs and establish end-to-end traceability across all project artifacts.	Section 4: Requirements Traceability Matrix (RTM) & Peer Review Log
	Week 12	ERP Setup & Test Design	Configure core Odoo modules (Sales, Purchase, Inventory, Invoicing) with domain-specific SKUs and draft UAT scripts.	Section 3.3 (Draft): System Master Data Setup & UAT Test Cases Specification
	Week 13	Verification & Defect Logging	Execute peer UAT testing on configured Odoo environments and document execution outcomes.	Section 3.3 (Final): Executed UAT Test Log with Pass/Fail Results & Screenshots
	Weeks 14 – 15	Defense & Final Deliverable	Compile final documentation, present system capabilities, and perform live End-to-End Odoo demos.	Final Master BRD (Phase 1 + 2) & Live Odoo System Presentation

CRITICAL RULE FOR PHASE 1: Phase 1 evaluates **pure Business Analysis capabilities**. Screenshots, system configurations, or user manuals of Odoo are **NOT required and WILL NOT be graded** during Phase 1. Focus 100% on business problems, processes, and stakeholder needs.